Database Systems

Lab - 5

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SQL Code

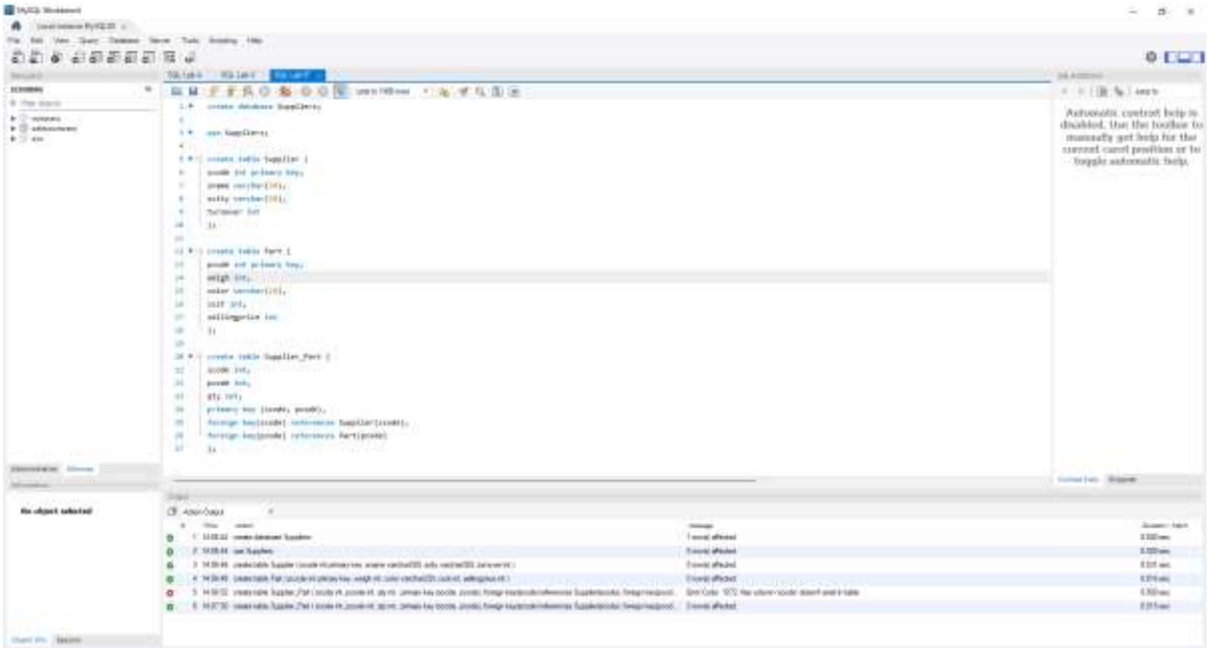
```
create database Suppliers;

use Suppliers;

create table Supplier (
scode int primary key,
sname varchar (50),
scity varchar (50),
turnover int
);

create table Part (
pcode int primary key,
weigh int,
color varchar (20),
cost int,
sellingprice int
);

create table Supplier_Part (
scode int,
pcode int,
qty int,
primary key (scode, pcode),
foreign key(scode) references Supplier(scode),
foreign key(pcode) references Part(pcode)
);
```



- The first step that I have performed was to create a new database by the name suppliers, which is visible on the left side of the windows.
- The use statement ensures that all the subsequent operations are executed inside this database.
- The next set of steps that I did was the creation of tables as per the requirement with the following set of constraints.
  - In the Supplier table the column **scode** must be a primary key.
  - In the Part table the column **pcode** must be a primary key.
  - In the **Supplier_Part** table the columns **sode** and **pcode** must be primary keys and **scode** and **pcode** of the **Supplier_Part** must be foreign keys and should get referenced from Supplier and Part tables respectively.

SQL Code

INSERT INTO Supplier (scode, sname, scity, turnover) VALUES

(101, 'Alpha Supplies', 'Bombay', 50),

(102, 'Beta Traders', 'Delhi', 80),

(103, 'Gamma Distributors', 'Bombay', NULL),

(104, 'Delta Corp', 'Chennai', 120),

(105, 'Epsilon Ltd', 'Pune', 200);

INSERT INTO Part (pcode, weigh, color, cost, sellingprice) VALUES

(1, 20, 'Red', 20, 25),

(2, 30, 'Blue', 30, 45),

(3, 35, 'Green', 40, 55),

(4, 40, 'Yellow', 25, 35),

(5, 50, 'Black', 60, 75);

INSERT INTO Supplier_Part (scode, pcode, qty) VALUES

(101, 2, 100),

(102, 1, 50),

(102, 3, 30),

(103, 2, 70),

(104, 4, 20),

(105, 5, 60);

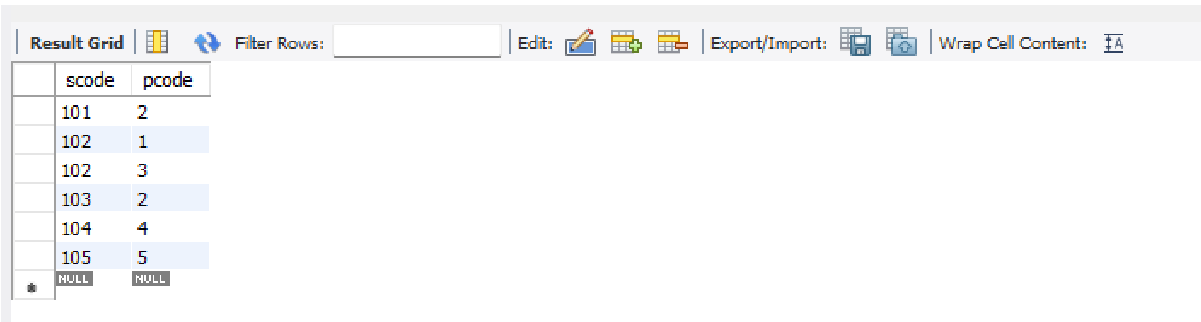
Write appropriate SQL Statement for the following

1. Get the supplier number and part number in ascending order of supplier number

```
-- Get the supplier number and part number in ascending order of supplier number
select scode, pcode from Supplier_Part order by scode, pcode asc;
```

SQL Code

select scode, pcode from Supplier_Part order by scode, pcode asc;



scode	pcode
101	2
102	1
102	3
103	2
104	4
105	5
NULL	NULL

2. Get the details of supplier who operate from Bombay with turnover 50

```sql
-- Get the details of supplier who operate from Bombay with turnover 50.
select * from Supplier where scity = "Bombay" and turnover = 50;
```

SQL Code

select * from Supplier where scity = "Bombay" and turnover = 50;

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

scode	sname	scity	turnover
101	Alpha Supplies	Bombay	50
NULL	NULL	NULL	NULL

3. Get the total number of supplier

```sql
-- Get the total number of supplier
select sum(qty) from Supplier_Part;
```

SQL Code

select sum(qty) from Supplier_Part;

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

sum(qty)
330

4. Get the part number weighing between 25 and 35.

```sql
-- Get the part number weighing between 25 and 35.
select pcode from Part where weigh between 25 and 35;
```

SQL Code

select pcode from Part where weigh between 25 and 35;

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

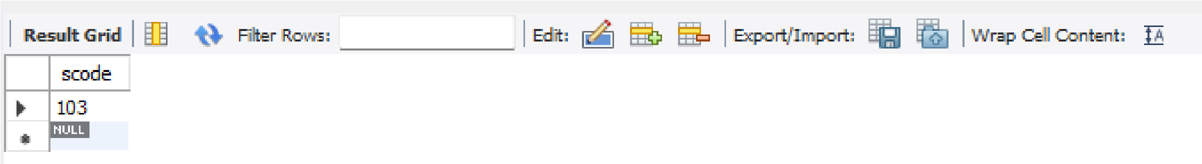
pcode
2
3
NULL

5. Get the supplier number whose turnover is null.

```
-- Get the supplier number whose turnover is null.
select scode from Supplier where turnover is null;
```

SQL Code

select scode from Supplier where turnover is null;

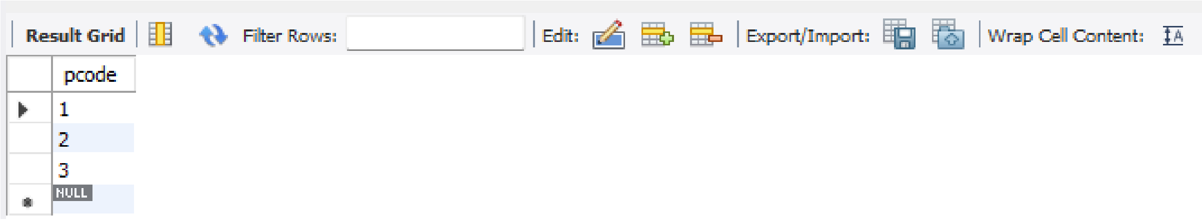


6. Get the part number that cost 20, 30 or 40 rupees

```
-- Get the part number that cost 20, 30 or 40 rupees
select pcode from Part where cost = 20 or cost = 30 or cost = 40;
```

SQL Code

select pcode from Part where cost = 20 or cost = 30 or cost = 40;

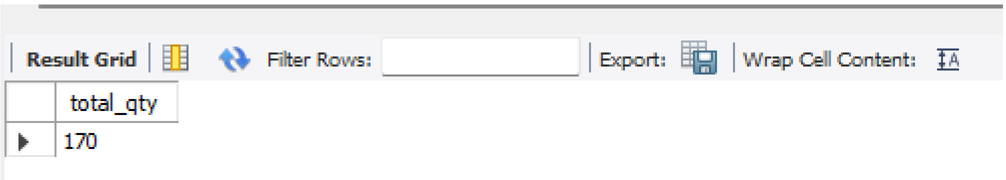


7. Get the total quantity of part 2 that is supplied.

```
-- Get the total quantity of part 2 that is supplied.
select sum(qty) as total_qty from supplier_part where pcode = 2;
```

SQL Code

select sum(qty) as total_qty from supplier_part where pcode = 2;

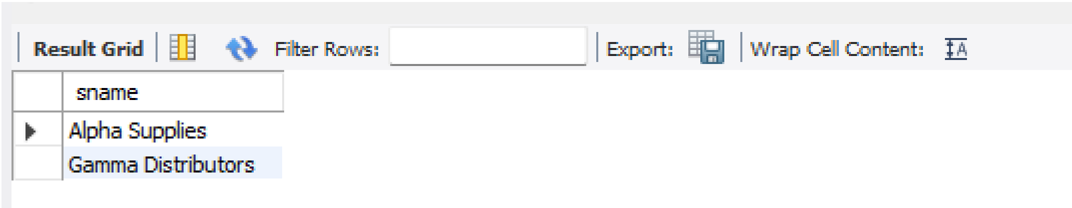


8. Get the name of supplier who supply part 2

```sql
--  Get the name of supplier who supply part 2
select sname from supplier where scode in (select scode from supplier_part where pcode = 2);
```

SQL Code

select sname from supplier where scode in (select scode from supplier_part where pcode = 2);



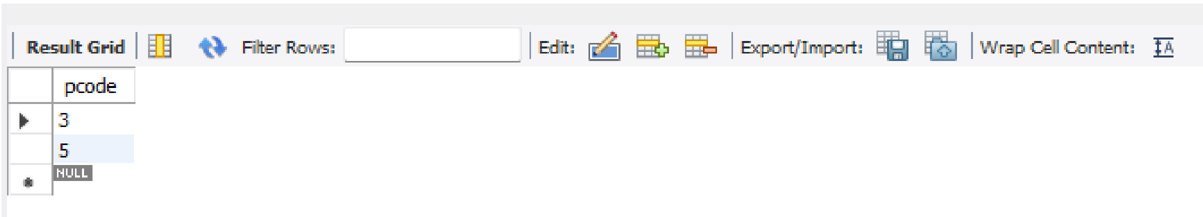
sname
Alpha Supplies
Gamma Distributors

9. Get the part number whose cost is greater than the average cost

```sql
-- Get the part number whose cost is greater than the average cost
select pcode from part where cost > (select avg(cost) from part);
```

SQL Code

select pcode from part where cost > (select avg(cost) from part);



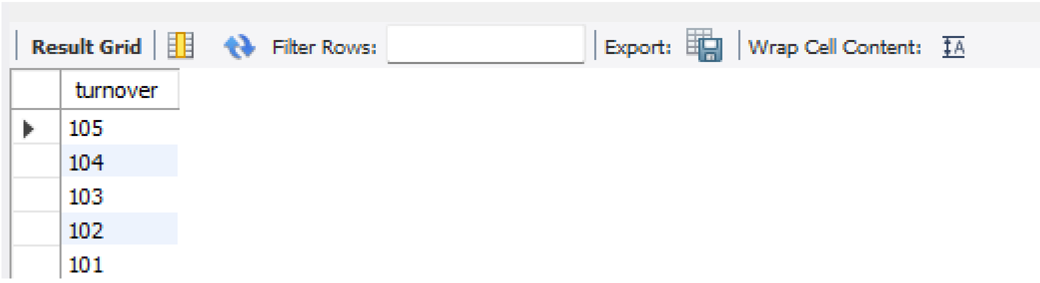
pcode
3
5
NULL

10. Get the supplier number and turnover in descending order of turnover.

```
-- Get the supplier number and turnover in descending order of turnover.
select scode turnover from supplier order by turnover desc;
```

SQL Code

select scode turnover from supplier order by turnover desc;



turnover
105
104
103
102
101